

Customer Churn Analysis

Data Analyst Report | Source: supplied Churn_Analysis notebook



GitHub: [surajitaiml711-cloud](#)

Customers	Churn	Retention	ARPU	Revenue at Risk
21	28.57%	71.43%	■18.85	■73.94K
Escalation Rate	Avg Complaints/User	Avg Tenure	Escalation ↔ Churn	Focus
19.05%	0.43	1,539 days	0.77	Segmented retention

Executive Summary

The supplied notebook analyzes 21 customers by combining customer profile, subscription and support data. The overall churn rate is 28.57%, with 71.43% retained. Churn is highly uneven across segments: the Basic plan records 60.00% churn, while Premium records 14.29%. The notebook also reports a 0.77 correlation between support escalations and churn, making escalated interactions a useful candidate for retention monitoring. Because the sample is small, these findings should be treated as directional and validated with additional data.

1. Data Sources & Preparation

The notebook extracts three SQLite tables: db_customer, db_subscription and db_support. Customer fields are cleaned by renaming name to customer_name, dropping interests and pincode, converting DOB to datetime, standardizing gender labels, and filling missing country values using state information. Subscription dates are converted to datetime and churn_flag is derived from cancellation_date. Support fields remove col_1 and comment; complaint_count is calculated before the latest support record per customer is retained for the final merge.

The final merged analytical dataset contains 21 rows. The support table initially has 9 records across 7 unique customers, so deduplication is important before the customer-level merge.

2. Core Churn KPIs

Metric	Notebook result
Churn rate	28.57%
Retention rate	71.43%
ARPU / average monthly charge	■18.85
Revenue at risk	■73.94K
Escalation rate	19.05%
Average complaints per user	0.43
Average tenure	1,539 days

3. Segment Findings

Plan-level churn

Plan	Customers	Churn rate
Basic	5	60.00%
Premium	7	14.29%
Standard	9	22.22%

Basic is the highest-risk plan in this sample, with 3 churned customers out of 5. Premium has 1 churned customer out of 7, while Standard has 2 out of 9.

Subscription-type churn

Subscription type	Customers	Churn rate
Organic	9	0.00%
Paid	6	16.67%
Refferal	6	83.33%

The source field is spelled Refferal. Its 83.33% churn rate is notable but is based on only six customers, so it should be validated before making acquisition-policy decisions.

4. Visual Findings

Churn Rate by Plan Type

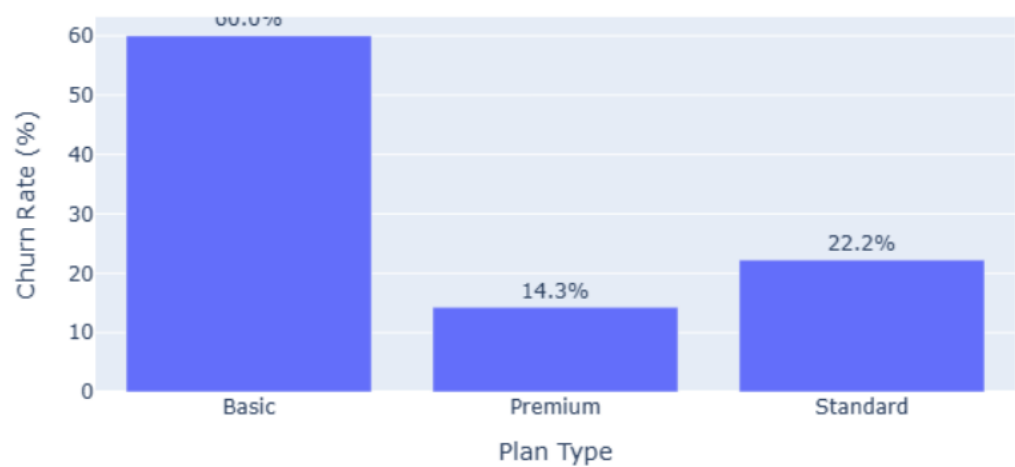


Figure 1. Churn Rate by Plan Type

Monthly Churn Trend



Figure 2. Monthly Churn Trend

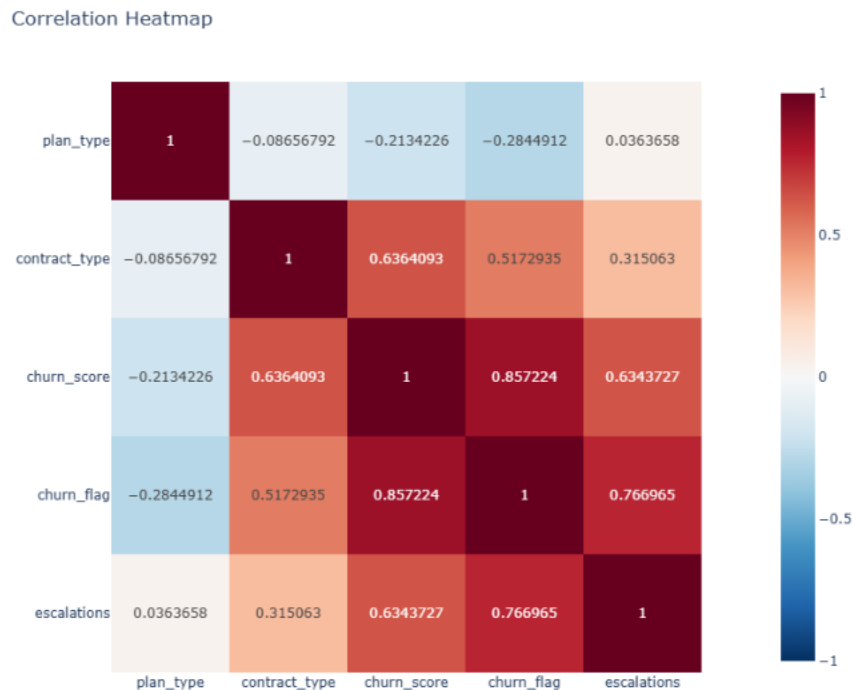


Figure 3. Correlation Heatmap

5. Correlation & Statistical Interpretation

The notebook reports a 0.77 correlation between the binary escalation indicator and churn_flag. The correlation heatmap also shows strong positive relationships between churn_score, churn_flag and escalations. These values indicate association in the observed data, not proof that escalations cause churn.

The notebook encodes plan type as Basic=0, Standard=1, Premium=2 and contract type as Monthly=0, Annual=1 before calculating correlations. Because these are categorical variables represented by ordered numeric codes, their correlation coefficients should be interpreted cautiously.

6. Business Recommendations

Prioritize Basic-plan retention: Review pricing, product value, onboarding and cancellation reasons for the Basic segment.

Create an escalation-based alert: Flag escalated support cases for proactive retention outreach, while validating the relationship on a larger sample.

Audit referral acquisition: Investigate the high churn observed among Referral customers and validate with more acquisition cohorts.

Build a customer risk score: Combine churn_score, plan, contract, tenure, support activity and CLTV to prioritize interventions.

Track revenue exposure: Monitor revenue at risk alongside customer counts so retention efforts focus on commercially important accounts.

7. Limitations & Data Quality

- The analysis contains only 21 customers, making percentage estimates sensitive to individual records.
- Several state-level groups are very small; extreme rates such as 100% may represent only a couple of customers.

- The escalation/churn correlation is not causal evidence.
- Customer age is calculated using the current year in the notebook rather than an exact birthday-adjusted age.
- The support table contains repeated customer records; the notebook calculates `complaint_count` and then retains the latest support row per customer for the final merge.
- The notebook contains a separate SQLite practice section with a users table; it is not part of the churn dataset analysis.

8. Analyst Conclusion

The analysis points to concentrated churn risk: Basic-plan customers, Referral customers and customers involved in escalated support interactions deserve particular attention. The most useful next step is not to over-generalize from the 21-customer sample, but to validate the patterns on a larger customer population and operationalize a customer-level retention dashboard or churn-risk model.

Prepared from the supplied Churn_Analysis notebook and its reported outputs.